Daniel (Chenyu) Lin

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EDUCATION

Cornell University **Ithaca, NY**
B.A. in Computer Science and Mathematics (double major;Honors in progress) Aug 2024 – Expected May 2027

University of North Carolina at Chapel Hill **Chapel Hill, NC**
Transferred Out, Computer Science Aug 2023 – May 2024

Selected Coursework:
AI/ML: Machine Learning | Large-Scale ML| Natural Language Processing | Reinforcement Learning (ongoing)
Maths: Information Theory (grad) | Measure Theory (grad, ongoing) | Stochastic Processes | (Advanced) Linear Algebra | Honors Analysis
General CS: Algorithms | Functional Programming | Data Structures | Embedded Systems

RESEARCH INTERESTS

I am interested in developing **multimodal learning methods that uncover underlying structure and relationships between modalities with limited human annotation**, especially in audio and music, where paired data are scarce. My methodological focus is on generative modeling, and I seek to combine mathematical analysis with empirical evaluation to understand what models can learn with limited supervision and develop reliable ways to assess their capabilities.

PUBLICATIONS

- **Daniel Chenyu Lin**, Michael Freeman, John Thickstun. "Assessing Factual Music Comprehension in Large Audio Language Models." *International Society for Music Information Retrieval (ISMIR), 2026*.
- Louis Bradshaw, Irmak Bukey, **Daniel Chenyu Lin**, John Thickstun, Chris Donahue. "PianoBench: Evaluating Musical Perception and Reasoning in Language Models." *In submission to ICLR 2027*.
- Saebyeol Shin, Chao Wan, Zhenzhen Liu, Justin Lovelace, **Daniel C. Lin**, Kilian Q. Weinberger, John Thickstun. "Music Transcription with (Almost) No Supervision." *Preprint.*

RESEARCH EXPERIENCE

Cornell Bowers CIS | Advisor: Dr. John Thickstun **Ithaca, NY**
Student Researcher, Generative Models & Music Information Retrieval Oct 2024 – Present

- **Factual evaluation of large audio language models** *Oct 2024 – Apr 2026*
  Showed that standard text-similarity evaluation on MusicQA cannot distinguish LALM responses conditioned on correct audio from responses conditioned on random audio. Designed a factual evaluation protocol that parses open-ended responses into structured labels scored by precision/recall/F1, and benchmarked nine frontier and open-source models across six factual retrieval tasks on MusicNet, FMA, and a newly curated dataset. Awarded a $6,000 research grant from Cornell's College of Arts and Sciences. (First author; accepted to ISMIR 2026)
- **Unsupervised automatic music transcription** | Co-advisor: Dr. Kilian Q. Weinberger *Oct 2025 – May 2026*
  Co-authored a study of unsupervised transcription with light supervision to break the symmetry on symbolic marginals; built the training corpus, scraping and normalizing ten sources into ~440 hours of audio and ~57 hours of symbolic data across four annotation formats. (Preprint)
- **Symmetry and identifiability in unsupervised translation** *May 2026 – Aug 2026*
  Led the theoretical analysis of why unsupervised transcription recovers the target map up to a constant pitch shift, characterizing the symmetry group of the symbolic marginal and deriving deterministic and variational cycle-consistent objectives with gradient penalties for a controlled 2D translation testbed, then analyzing why these objectives fail to isolate the intended solution.
- **Instruction tuning LLMs for MIDI understanding** | Co-advisor: Dr. Chris Donahue *Aug 2026 – Present*
  Built an instruction-tuning pipeline supporting full fine-tuning, LoRA, and DoRA for LLMs following symbolic music mid-training. Evaluated music understanding, general knowledge, and instruction following using PianoBench, MMLU-Redux, and IFEval. Combined music mid-training and instruction tuning increased PianoBench overall multiple-choice accuracy from 42.8% to 56.0%. Instruction tuning recovered losses in general knowledge and instruction following while largely preserving musical understanding. (In submission to ICLR 2027)

Lyra Lab, Tencent Music Entertainment **Shenzhen, China**
Research Intern, Audio Information Retrieval Jun 2024 – Aug 2024

- Replaced the Conformer encoder in a production Cover Song Identification system with a Zipformer encoder, raising top-1 accuracy by 7 points to over 90%, and resolved exploding gradients at scale by reimplementing the block's activation function.
- Pre-trained MT3 from scratch on roughly twice the data used in the original paper, improving instrument detection and onset alignment.
- Built an in-batch audio augmentation pipeline supporting speed and pitch variation, noise injection, and impulse response convolution.

Music Tech X (High School Program) Student Researcher & Group Leader Jun 2023 – Aug 2023

- Implemented an audio fingerprinting algorithm from scratch for music recognition, and led a team of six in improving its search and evaluation components to reach 96% precision on the MIREX dataset.
- Explored a learned alternative to fingerprinting, training a ResNet to recognize pitch-shifted and tempo-varied music with over 80% precision.

TEACHING & ACADEMIC INVOLVEMENT

Reading Group Participant & Presenter
Cornell University & Ithaca College **Ithaca, NY**

- Theory of Generative Modeling (CS 7792) Spring 2026
- Music Technology Fall 2024 – Spring 2025

Teaching Assistant
Cornell Bowers CIS & Cornell College of Engineering **Ithaca, NY**

- *ECE 5620/ECE 7620: Information Theory and Generative Modeling Fall 2026
- CS 4782/CS 5782: Introduction to Deep Learning Spring 2026
- CS 3780/CS 5780: Introduction to Machine Learning Fall 2025
- *CS 1109: Foundational Programming Concepts Summer 2025

  * denotes sole TAship for the course.

TECHNICAL SKILLS

Programming Languages: Python, Java, OCaml, C (cuBLAS), ARM Assembly
Machine Learning: Pytorch, Tensorflow, Huggingface, W&B, Data Augmentations
Audio Programming: Librosa, Torchaudio, Madmom, Musescore, Logic, Audacity
Other Useful Skills: Linux, Slurm Cluster, GPU Deployment, Github, Google Colab